

Living Work — Caring for Country

Field notes format. Three layers: the work, infrastructure and technology, and what the work produces.

Field notes from Turraburra

YACHATDAC manages Turraburra, 8,870 hectares of Iningai Country 120km north of Barcaldine in Central Western Queensland. The property was bought back for the Iningai people in 2019.

Iningai Rangers deliver the work: fire-stick farming, land restoration and management, wildlife conservation and site protection, on a property that runs on its own water, power and communications.

Our challenges

[Expandable list. Each opens to context and what is being done.]

A century of overgrazing

Turraburra was run as a sheep and cattle station from the 1880s. Continuous grazing strips groundcover, compacts soil and stops native grasses setting seed. Fencing lets us rest paddocks; seed collection and replanting rebuild what grazing removed.

Degraded springs and waterholes

Springs are the most valuable water here and the first thing stock destroy. Restoration is slow: clear the hole, plant around it, keep water up to it through the first dry.

Soil carbon loss and erosion

Testing here found carbon beginning only 30 centimetres down. Groundcover, right-way fire and rested paddocks build it back. Worth measuring before you start work rather than after.

Wrong-way fire and wildfire risk

Without regular cool burning, fuel builds and the fire that comes is hot enough to kill trees, sterilise soil and damage sites. Cool burns break Country into a patchwork.

Weeds and feral animals

Ferals foul waterholes, eat regrowth and damage rock shelters. Weeds move in wherever ground is disturbed. Control is continuous and targeted.

Extreme heat, drought and flood

The highest-likelihood risk on this property. Workplans are built loose enough to move around the season.

Distance from town

Barcaldine is 120km south, Aramac the last town. Everything carries freight cost and waiting time. Machinery is serviced and rebuilt on site.

No reliable phone reception

No dependable mobile coverage apart from patches on the escarpment. A safety issue before it is an inconvenience. See Infrastructure.

Water security across 8,870 hectares

Stock, camp, nursery and firefighting water all come off one system. A standing priority rather than a project with an end date.

Damage and access to cultural sites

Marra Wonga carries graffiti going back over a century, and artefacts have been removed since the late 1800s. Roads go around sites; access is guided.

Species we have not yet identified

You cannot protect or claim credit for what you have not recorded. Acoustic recorders and cameras are building that record.

Ranger funding ahead of income

Grant cycles are shorter than the work. Losing ranger funding before commercial income is established is the single biggest risk to everything here.

Natural capital market volatility

Prices move and methodologies change. The approach is several natural capital products rather than one.

Iningai Rangers

Rangers run the cool burns, the fencing, the seed collection, the spring and wetland restoration, the weed and feral control, the machinery and the site protection. Ranger workplans set the year's work across the property. Training partnerships are being built so the jobs here can be filled from community.

[Rotating carousel of Ranger photographs, auto-advancing, with captions. Replaces the earlier text panels.]

The work

What Rangers do on the ground. Alternating photograph and text.

01

Fire-stick farming

Low-intensity cool burning at the right time of year — reducing fuel loads, encouraging germination and breaking Country into a patchwork that carries more life than unburnt or hot-burnt ground.

Rangers burn right way and look for black ash rather than scorched ground. Burns run as workshops as well as work. Soil is tested before and after: carbon here began only 30 centimetres down, and one paddock was recorded working as a carbon sink within eight to nine months of a cool burn.

02

Springs and waterholes

Bringing cultural wetlands and spring networks back to holding water on their own.

One spring had been trampled flat by a century of stock. Rangers cleaned the waterhole out, planted grasses around it and carted a thousand litres to it twice a day through drought. Eight days in it held a puddle. The next morning there was a koala standing at it.

03

Seed and groundcover

Collecting seed when Country produces it, holding it, and putting it back into revegetation.

Seed is available when the season decides, not when a work plan says so, which is why the year is organised around the country rather than the other way round.

04

Fencing and paddocks

New fencing built corner assembly by corner assembly, giving paddocks that can be managed separately.

Rangers built it themselves, learning the job as it went. Without separate paddocks the flux tower comparison would not have been possible.

05

Weeds and feral animals

Continuous control targeted at the places doing the most damage — waterholes, regrowth and rock shelters.

Control is ongoing rather than a one-off campaign, and runs alongside the restoration work.

06

Sites and access

Protecting Marra Wonga and the other recorded sites, and controlling how people reach them.

Road alignments were set from cultural mapping so routes work around sites rather than through them. Ecological and cultural mapping tells us what is where and what must not be disturbed.

07

Monitoring

Recording what is here and what changes, so management decisions and credit claims both rest on evidence.

Four acoustic recorders log birds and frogs into a database held with QUT, two of them at the springs. Cameras on the towers photograph five hundred metres around. Soil samples from the bore are held by QUT.

Infrastructure and technology

What it takes to run a property 120 kilometres from the nearest town.

[Dark section. Schematic diagram — Starlink to router to mesh; UHF coverage radius; bore to tanks, camp and stock; solar feeding the monitoring gear. No photographs.]

COMMUNICATIONS Starlink uplink bridged to an external router · wifi mesh with three extenders plus one in the campground · UHF radio in the house, the shed and nearly every vehicle · about ten UHF handhelds, roughly 5km range from the house

WATER 480-metre bore at Lancewood Ridge · tanks, pumps and lines across the property · filters on a six-month replacement cycle · stock, camp, nursery and firefighting water off one system

POWER Solar with wifi-enabled regulators · data can be pulled off the regulators remotely · 240 volt supply to the mesh network

MONITORING GEAR Two flux towers — managed Country and grazed control · four acoustic recorders, two at the springs · cameras on the towers, 500m radius · soil samples every six metres from the bore, held with QUT

MACHINERY Grader, buggies, vehicles and heavy plant · serviced, repaired and rebuilt on site · register down to serial and filter numbers · service history with hours and kilometres, updated every visit

COMING Secure Ranger Base · research facilities and nurseries · walking tracks and camping facilities · staff housing

What the work produces

The same activities, measured.

REGISTRATION UNDERWAY Carbon Two flux towers and soil testing underpin a carbon farming project. Cool burning done right way is the practice being measured.	BUILDING THE RECORD Biodiversity credits Species of significance are being identified and monitored. The same data that guides management underpins any credit work.	IN PROGRESS IPA designation Indigenous Protected Area status for Turraburra, alongside National and World Heritage applications and the Four Rivers People Native Title claim.
BEING DEVELOPED Fee-for-service Ranger skills offered to neighbouring properties — right-way fire, land management and cultural heritage advice.	[FOR YACHATDAC TO WRITE] Rainbow Credits Integrating environmental and traditional land management into a single natural-capital product. Needs describing in your words: what it is, what is measured, who it is for, and where it sits with the Rainbow Foundation.	

[Status labels to be confirmed before publishing.]

GET INVOLVED

Ancient traditions walking together with contemporary visions

[Full-bleed band — photograph behind, dark scrim, three pathways side by side, email signup beneath.]

Ranger exchange On-Country training camps and exchanges with other First Nations ranger groups. Get in touch →	Fund the work Ranger wages, water infrastructure, monitoring equipment and restoration. What is needed is specific and can be costed. Partner with us →	Land management services Right-way fire, cultural heritage advice and Country management for properties in the district. Enquire →
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[Consider a fourth card for donations if DGR status is in place.]